

Reduce Cost

SUMMARY

Current architecture: A new container is created per assessment to guarantee patient data isolation, a safe pragmatic call, but expensive. Each cold start loads the ML model from scratch.

At low volumes this is manageable, it becomes a commercial problem as we scale.

TECH STACK

Current - Flask

Target - AKS · Azure Service Bus · FastAPI

APPROACH & RECOMMENDATION

Refactor to a stateless inference service enabling a warm pool where model weights stay loaded in memory. Patient isolation is guaranteed by design, not container boundary. Async queue (Azure Service Bus) absorbs demand spikes.

On-device signal extraction moves face detection, ROI extraction, and POS + CHROM processing to the device. Only the processed BVP signal is transmitted. Removes a further chunk of cloud compute per assessment.

ALTERNATIVES CONSIDERED & REJECTED

Serverless inference (Azure Functions) the cold start still happens, it's just hidden. Full stateless rewrite is the right destination but too risky to do in one go on a live clinical system. Incremental refactor is safer.

Improve Performance — Cloud & Client

SUMMARY

The phone is a signal processor, not just a camera. Performance improvements come from two places: better signal quality captured on-device, and faster inference in the cloud. Together they reduce total assessment time and failed assessments.

TECH STACK

Android - MediaPipe · Vulkan · Kotlin
iOS - Vision framework · Metal · Swift/SwiftUI

APPROACH & RECOMMENDATION

Client: Metal (iOS) and Vulkan (Android) GPU frameworks run the signal extraction pipeline on-device. MediaPipe tracks face landmarks continuously. ROI extraction targets forehead and cheeks. POS + CHROM fusion extracts the blood volume pulse signal. A signal quality gate rejects low-quality frames before transmission, only clean signal reaches the cloud.

Cloud: warm pool means inference begins the moment signal arrives, no cold start wait. Cleaner incoming signal means the model runs faster with higher confidence.

The ISP (image signal processor) has auto-exposure, auto-white-balance, H.264 compression which degrades the rPPG signal. The on-device pipeline is designed to work around this.

ALTERNATIVES CONSIDERED & REJECTED

All compute in cloud, raw video transmitted, simpler day one but slower, phase 1 reality while stateless refactor is in progress. Shared C++ core across iOS and Android, reduces duplication but sacrifices platform-native performance.